

A method of three-dimensional object tracking in a video with separation into static and dynamic elements

Motivation. Vision-language large models (VLLMs) increasingly require rich spatial-temporal context to understand dynamic scenes and plan actions Yang et al. (2025). Recent work on Set-of-Mark (SoM) and Trace-of-Mark (ToM) Yang et al. (2025) demonstrates that marking actionable objects and tracing their movements provides crucial visual prompts for agentic tasks. However, a critical gap remains: distinguishing between static scene elements and genuinely dynamic objects in videos with camera motion, and providing accurate 3D trajectories as structured prompts for downstream VLLMs. Existing 2D tracking methods Wojke et al. (2017); Pang et al. (2021); Li et al. (2023) cannot resolve this ambiguity, while recent 3D reconstruction approaches Zhuo et al. (2025); Lan et al. (2025) focus on geometry without object-level motion analysis. We propose **ToM++** (*Trace of Marks++*), a framework that explicitly classifies objects as static or dynamic through 3D motion analysis with camera compensation, and generates comprehensive 2D/3D traces that serve as actionable prompts for VLLMs in embodied AI tasks. The method is particularly promising for applications in self-driving cars, human/robot indoor orientation, and object manipulation (our domain).

Method Overview. ToM++ processes monocular video through three main stages: (1) video pre-processing with temporal subsampling and spatial rescaling, (2) patch-based online processing integrating 3D reconstruction, semantic segmentation, and 2D point tracking, and (3) statistical motion analysis with camera motion compensation to classify objects as static or dynamic. The method operates in

an online streaming fashion, processing video patches incrementally while maintaining spatial and temporal consistency across the entire sequence. Figure 1 illustrates the overall pipeline.

Input and Preprocessing. Given an input video $\mathbf{V} \in R^{N \times H_{\text{orig}} \times W_{\text{orig}} \times 3}$ with N frames at resolution $(H_{\text{orig}}, W_{\text{orig}})$, we apply temporal subsampling with step size s and spatial rescaling such that $\max(H, W) = 700$ pixels, producing a processed video $\mathbf{V}' \in R^{N' \times H \times W \times 3}$ where $N' = \lfloor N/s \rfloor$. This resolution balances computational efficiency with reconstruction quality. The preprocessed video is then divided into overlapping patches of size b (typically 5 frames) with 1-frame overlap to ensure temporal continuity.

Streaming 3D Reconstruction. For each video patch, we employ StreamVGGT Zhuo et al. (2025), a streaming 4D visual geometry transformer that reconstructs dense spatial-temporal geometry using causal attention with KV-cache for efficient online processing. The KV-cache mechanism enables significant speedup, achieving up to 2.5 fps on Nvidia A100 (including CPU computations), which already allows the method to be used in online tasks. An additional benefit is the possibility to run the method on long videos on consumer-grade video cards such as Nvidia RTX 4060Ti 16GB, making the approach accessible for real-world deployment. StreamVGGT produces multiple outputs for each processed frame:

- RGB images: $\mathbf{I} \in R^{N' \times 3 \times H \times W}$
- Dense 3D world points: $\mathbf{P}_{\text{world}} \in R^{N' \times H \times W \times 3}$ with confidence $\mathbf{C}_{\text{world}} \in R^{N' \times H \times W}$

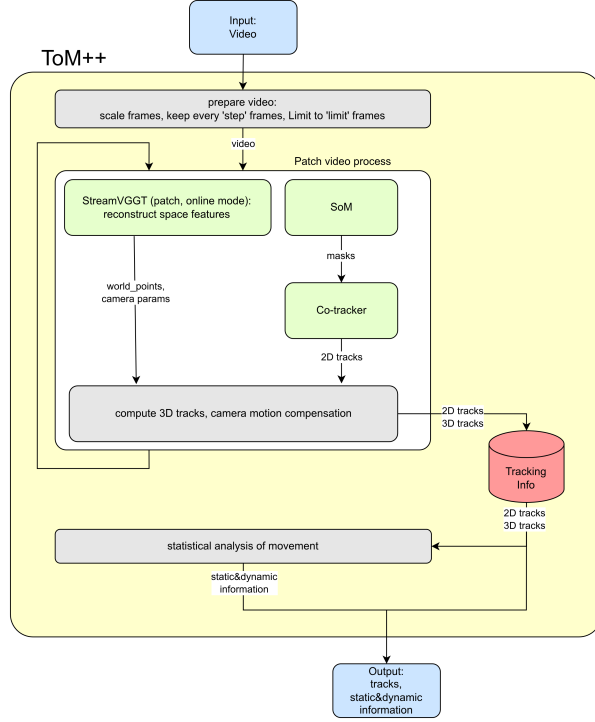


Figure 1: ToM++ method overview: illustrating the three main processing stages and data flow through the pipeline.

- Depth maps: $\mathbf{D} \in R^{N' \times H \times W \times 1}$ with confidence $\mathbf{C}_{\text{depth}} \in R^{N' \times H \times W}$
- Pose encoding: $\mathbf{E}_{\text{pose}} \in R^{N' \times 9}$, which we then transform into camera extrinsics $[\mathbf{R}|\mathbf{t}] \in R^{N' \times 3 \times 4}$ and intrinsics $\mathbf{K} \in R^{N' \times 3 \times 3}$ for subsequent 3D projection operations.

The causal attention mechanism with KV-cache enables StreamVGGT to maintain temporal consistency while processing frames incrementally, making it suitable for streaming scenarios.

Semantic Segmentation with SoM. At the first frame of each patch, we apply our Set of Marks (SoM) module to detect and segment object instances. SoM leverages YoloE Cheng et al. (2024) in

open-vocabulary mode with a custom dictionary, enabling flexible object detection across diverse scenes without retraining. The model generates instance masks $\mathbf{m}_j \in \{0, 1\}^{H_{\text{orig}} \times W_{\text{orig}}}$ for each detected object j . For each mask, we extract:

- Bounding box: $\mathbf{b}_j \in R^4$ (format: $[x_1, y_1, x_2, y_2]$)
- Feature embedding: $\mathbf{e}_j \in R^{512}$ from YoloE’s intermediate features
- Mask center: $\mathbf{c}_j \in R^2$

The embeddings enable object re-identification across patches through similarity matching, supporting long-term tracking consistency.

2D Point Tracking. To establish correspondences within each patch, we employ CoTracker2 Karaev et al. (2023), a transformer-based point tracker. We initialize a dense grid of M tracking points (with grid size typically 100) on the first frame, then filter these points to associate them with detected object masks. The patch-based processing strategy is particularly advantageous for CoTracker2: the small temporal window and minimal inter-frame changes within patches significantly reduce tracking failures, while the model’s efficiency enables fast processing of short sequences. CoTracker2 produces 2D trajectories $\mathbf{T}_{2D} \in R^{1 \times (b+1) \times M \times 2}$ and visibility predictions $\mathbf{V}_{\text{vis}} \in \{0, 1\}^{1 \times (b+1) \times M}$ for all tracked points across the patch. For objects appearing in subsequent patches, we match them with previously tracked objects using mask embeddings and spatial overlap, enabling query point reinitialization at the last known 2D positions for temporal continuity.

3D Object Tracking with Camera Compensation. We lift 2D point tracks to 3D world coordinates using the camera parameters and depth maps from StreamVGGT. For each object i and frame t , we project the 2D track position $\mathbf{T}_{2D}[t, i] = (u, v)$ to 3D:

$$\mathbf{T}_{3D}[i, t] = \mathbf{R}_t^{-1} \left(d(u, v) \cdot \mathbf{K}_t^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} - \mathbf{t}_t \right), \quad (1)$$

where $d(u, v)$ is the depth at pixel (u, v) , and $[\mathbf{R}_t | \mathbf{t}_t] = \mathbf{R}_t$ is the camera extrinsic matrix. This produces 3D object trajectories $\mathbf{T}_{3D} \in R^{L \times N' \times 3}$ for L unique objects in world coordinates, effectively compensating for camera motion.

We aggregate tracking information across all patches, maintaining a *bag of objects* consisting of:

- Object IDs: $\mathcal{I} = \{1, 2, \dots, L\}$
- Object embeddings: $\mathbf{E}_{\text{obj}} \in R^{L \times 512}$
- 3D point clouds: $\mathbf{C}_{3D} \in R^{L \times 1000 \times 3}$ (sampled from masked depth)

Statistical Motion Analysis. To classify objects as static or dynamic, we compute the total displacement for each object across the video. We apply a two-threshold strategy: a local threshold $\varepsilon_{\text{local}}$ (default: 1/fps) filters out tracking noise and small jitter between consecutive frames, while a global threshold $\varepsilon_{\text{global}}$ determines the overall motion magnitude.

The total displacement is $D^i = \sum_{t=1}^{N'-1} \Delta_t^i$. Objects with $D^i \geq \varepsilon_{\text{global}}$ are classified as dynamic ($\mathcal{I}_{\text{pos}}$), while those with $D^i < \varepsilon_{\text{global}}$ are classified as static ($\mathcal{I}_{\text{neg}}$).

Output Representation. ToM++ outputs comprehensive tracking information including:

- Per-frame 2D/3D tracks stored
- Object projections: $\mathbf{P}_{\text{proj}} \in R^{N' \times N' \times L \times 2}$ projecting all objects from all source frames to all target frames using camera parameters, enabling trajectory visualization and downstream analysis
- Motion classification: $\mathcal{I}_{\text{pos}}$ (dynamic) and $\mathcal{I}_{\text{neg}}$ (static)
- Object metadata: class labels, embeddings, and 3D point clouds

This rich representation supports diverse downstream applications including scene understanding and object manipulation planning.

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